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# Bulletin

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**Classification of Liquids for the Approval of Liquid Meters**

## 1.0 Purpose

The purpose of this bulletin is to facilitate the approval of liquid meters that are designed to measure a variety of products. The various liquids normally measured volumetrically have been categorized into groups of liquids having similar physical characteristics and chemical compatibility.

## 2.0 Scope

This bulletin is applicable to positive displacement meters, turbine meters with certain limitations on viscosity for a given group, and to mass flow meters.

## 3.0 Background

When meters are submitted for approval, the manufacturer is not always aware of all the potential products that the meter may be used to measure. Experience has shown that liquids can be grouped together according to characteristics that effect their measurability, such as viscosity, lubricity and compatibility. Prior to the development of this bulletin, a formal evaluation was required when requests for new products applications were submitted.

The tables found in this document have been developed in conjunction with the National Conference of Weights and Measures (NCWM) and are intended to be the same as those found in the NCWM Publication 14. This was done for the benefit of meter manufacturers.

## 4.0 Definitions

All words and expressions used in this bulletin have the same meaning as in the Regulations.

## 5.0 Use of the tables

**5.1** Approval applicants will initially request the types of products and the viscosity and density ranges they wish to be included in the Notice of Approval for a particular meter. The following sections detail how additional products will be added to approvals.

### 5.2 Positive Displacement and Turbine Meters

**5.2.1** A liquid meter that is successfully tested on only one liquid will be approved for use in trade for that liquid only.

**5.2.2** If subsequently, the meter is successfully tested on a second liquid in the same sub-group, the meter will be approved for the whole liquid sub-group provided that the viscosities of the two liquids involved adequately bracket the sub-group's range from the lowest to the highest viscosity.

**5.2.3** If an approval applicant finds a new product application for a meter that has already received approval, and the new product falls within the sub-group currently listed on the Notice of Approval, no additional approval testing is required. If the product does not fall within the sub-group currently listed, the approval applicant may request additional approval testing to have the new product added to the list of approved products.

### 5.3 Mass Flow Meters (Coriolis Meters)

**5.3.1** For a mass flow meter to receive approval for a single, normally liquid product, the density of the product should not vary by more than ±100 kg/m$^3$.

**5.3.2** For a mass flow meter to receive approval for a range of densities in the normal liquid product group, tests should be conducted with two liquids within the product group. Upon successful completion of these tests, all other normally liquid products having densities between the densities of the two products tested will be included in the Notice of Approval. This does not apply to the following product groups: Compressed Liquids, Cryogenic Liquids and Heated Products. In the case of these product groups, only one liquid within each group needs to be tested. Upon successful testing, the entire product group will be covered in the Notice of Approval.

**5.3.3** Multi-product applications (that is, applications in which the meter will be used to measure different products that have densities varying by more than 100 kg/m$^3$ without a change to zero or calibration of the meter) will include a multi-product test. During this test, the meter will be tested initially on one product of a known density and tested again on another product with a known density of at least ±200 kg/m$^3$ from the initial product density. The range may be extended by testing on a product from any of the product groups having a density greater than ±200 kg/m$^3$ from the initial product density.

## 6.0 Revisions

**6.1** Revision 1, issued 1996-01-01, has changes to the viscosity values to include units in SSU, centistokes and centipoise as well as slight changes to the specific gravity values.

**6.2** Revision 2, issued 2005-05-13, provides the policy for evaluating mass flow meters for different product applications. The bulletin was also reformatted and the "background" section added, along with a few other non-significant test changes.

**6.3** Revision 3 was intended to simplify requirements regarding classification of liquids for the approval of liquid meters. (PBRI2008)

## 7.0 Additional Information

For additional information regarding this bulletin, please contact the Senior Program Officer responsible for volume measurement. For more information regarding Measurement Canada and its programs, visit our Web site located at http://mc.ic.gc.ca/.


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**Table 1 - Families of Liquids as related to Positive displacement and Turbine Meters**

| PRODUCTS<br>PRODUITS | PRODUCT SUB-GROUPS<br>SOUS-GROUPES DES PRODUITS | EXAMPLES OF LIQUIDS WITHIN FAMILIES (**PARTIAL LISTING**)<br>EXEMPLES DE LIQUIDES D'UNE MÊME FAMILLE (**LISTE PARTIELLE**) | TYPICAL VISCOSITY<br>VISCOSITÉ RANGE<br>(SSU / *SSU*)<br>(centistokes)<br>(centipoise) | TYPICAL DENSITY @ 15°C<br>MASSE VOLUMI-QUE @ 15°C |
|---|---|---|---|---|
| PETROLEUM PRODUCTS<br>PRODUITS PÉTROLIERS<br>I | (A) REFINED PETROLEUM PRODUCTS<br>(A) PRODUITS PÉTROLIERS RAFFINÉS | Gasoline Essence<br>Kerosene Kérosène<br>Diesel Fuel Diesel<br>Distillate Distillat<br>Fuel Oil Mazout<br>Light Oil Huile légère<br>Spindle Oil Huile à broches | < 500<br>0.6 - 108<br>0.4 - 106 | 0.64 - 1.1 |
| | (B) AVIATION FUELS<br>(B) CARBURANTS AVIATION | AV gas Carburant aviation<br>Jet A Jet A<br>Jet A-1 Jet A-1<br>Jet B Jet B<br>JP4 JP4<br>JP5 JP5<br>JP7 JP7<br>JP8 JP8 | <40<br>0.6 - 4.2<br>0.4 - 3.6 | 0.65 - 0.85 |
| | (C) LUBRICATING OILS<br>(C) HUILES DE GRAISSAGE | SAE Grades Qualité S.A.E. | 150 - 10000<br>32 - 2200<br>28 - 2150 | 0.85 - 0.98 |
| | (D) HEATED PRODUCTS<br>(D) PRODUITS CHAUFFÉS | Bunker Oil Fuels de soutes<br>6 Oil Huile 6<br>Crude Oil Huiles brutes<br>Asphalt Asphalte | 150 - 10000<br>32 - 2200<br>25 - 2420 | 0.8 - 1.1 |
| SOLVENTS<br>SOLVANTS<br>II | (A) SOLVENTS - GENERAL<br>(A) SOLVANTS EN GÉNÉRAL | Acetates Acétates<br>Acetone Acétone<br>Esters Esters<br>Ethylacetate Acétate d'éthyle<br>Hexane Hexane<br>MEK MEK<br>Naphtha Naphta<br>Toluene Toluène<br>Xylene Xylène | < 35<br>0.6 - 2.8<br>0.4 - 2.5 | 0.6 - 0.90 |

| PRODUCTS<br>PRODUITS | PRODUCT SUB-GROUPS<br>SOUS-GROUPES DES PRODUITS | EXAMPLES OF LIQUIDS WITHIN FAMILIES (**PARTIAL LISTING**)<br>EXEMPLES DE LIQUIDES D'UNE MÊME FAMILLE (**LISTE PARTIELLE**) | TYPICAL VISCOSITY<br>*VISCOSITÉ*<br>RANGE<br>(SSU / SSU)<br>(centistokes)<br>(centipoise) | TYPICAL DENSITY @ 15°C<br>MASSE VOLUMIQUE @ 15°C |
|---|---|---|---|---|
| | (B) SOLVENTS - CHLORINATED<br>(B) SOLVANTS CHLORÉS | Carbon Tetra-chloride / Tétrachlorure de carbone<br>Methylene-chloride / Chlorure de méthylène<br>Perchloro-ethylene / Perchloroéthylène<br>Trichloro-ethylene / Trichloroéthylène | < 35<br>0.6 - 2.8<br>0.4 - 4.5 | 0.6 - 1.6 |
| | (C) ALCOHOLS, GLYCOLS & WATER MIXES THEREOF<br>(C) ALCOOLS, GLYCOLS ET LEURS MÉLANGES AQUEUX | Ethanol / Éthanol<br>Methanol / Méthanol<br>Butanol / Butanol<br>Isopropyl / Isopropyl<br>Isobutyl / Isobutyl<br>Ethylene glycol / Éthylène glycol<br>Propylene / Propylène<br>Glycol / Glycol | < 250<br>0.6 - 54<br>0.4 - 65 | 0.6 - 1.2 |
| COMPRESSED GASES LIQUEFIED GAZ LIQUÉFIÉ COMPRIMÉ<br>III | (A) LPG<br>(A) GPL | Propane / Propane<br>Butane / Butane<br>Freon 11 / Fréon 11<br>Freon 12 / Fréon 12<br>Freon 22 / Fréon 22 | n/a<br>0 .2 - 0.3<br>0.1 - 0.2 | 0.5 - 0.65 |
| | (B) NH3<br>(B) NH3 | Anhydrous-Ammonia / Ammoniac anhydre | n/a<br>0.5<br>0.4 | 0.62 |
| AGRICULTURAL LIQUIDS LIQUIDES AGRICOLES<br>IV | (A) FERTILIZERS - CLEAR LIQUID N-P-K<br>(A) ENGRAIS - LIQUIDE TRANSPARENT N-P-K | 10-34-0 / 10-34-0<br>4-10-10 / 4-10-10<br>9-18-9 / 9-18-9 | 50 - 100<br>7.5 - 20<br>5.3 - 26 | 0.7 - 1.3 |
| | (B) FERTILIZERS - NITROGEN SOLUTIONS<br>(B) ENGRAIS - SOLUTIONS AZOTÉES | 20% Aqua-Ammonia / 20% ammoniaque liquide<br>28%, 30% or 32% Nitrogen Solution / 28%, 30% ou 32% solution azotée<br>Urca / Urée<br>Ammonia-Nitrate / Nitrate d'ammonium | 32 - 100<br>1.5 - 20<br>1.1 - 27 | 0.7 - 1.35 |

| PRODUCTS<br>PRODUITS | PRODUCT SUB-GROUPS<br>SOUS-GROUPES DES PRODUITS | EXAMPLES OF LIQUIDS WITHIN FAMILIES (**PARTIAL LISTING**)<br>EXEMPLES DE LIQUIDES D'UNE MÊME FAMILLE (**LISTE PARTIELLE**) | TYPICAL VISCOSITY<br>VISCOSITÉ RANGE<br>(SSU / SSU)<br>(centistokes)<br>(centipoise) | TYPICAL DENSITY @ 15°C<br>MASSE VOLUMI-QUE @ 15°C |
|---|---|---|---|---|
| | (C) FERTILIZERS - SUSPENSIONS<br>(C) ENGRAIS - MATIÈRES EN SUSPENSION | 3-10-30 3-10-30<br>4-4-27 4-4-27 | 100 - 1000<br>20 - 215<br>18 - 360 | 0.9 - 1.65 |
| | (D) HERBICIDES - THIN LIQUIDS<br>(D) HERBICIDES - LIQUIDES MINCES | Eradicane Eradicane<br>Lorox Lorox<br>Princep Princep<br>Round-up Round-up<br>Sencor Sencor<br>Sutan Sutant<br>Sutazine Sutazine<br>Treflon Treflon | 100 - 250<br>20 - 54<br>18 - 65 | 0.6 - 1.2 |
| | (E) HERBICIDES - VISCOUS LIQUIDS<br>(E) HERBICIDES - LIQUIDES VISQUEUX | Dual Dual | 200 - 500<br>43 - 120<br>26 - 144 | 0.7- 1.2 |
| | (F) HERBICIDES - FLOWABLES<br>(F) HERBICIDES - FLUIDIFIABLES | Atrex Atrex<br>Atraxine Atraxine<br>Bicep Bicep<br>Bladex Bladex | 500 - 3500<br>118 - 760<br>106 - 912 | 0.9 - 1.2 |
| | (G) LIQUID FEEDS<br>(G) ENGRAIS LIQUIDES | Liquid-Molasses Mélasses liquides<br>Molasses + Phos-Acid and/or Urca Mélasses + Phos-acide et/ou urée | 50 - 250000<br>7.5 - 54,000<br>7.5 - 81000 | 1.0 - 1.5 |
| WATER<br>EAU<br>V | - | Tap Water Eau du robinet<br>De-ionized Déionisée<br>Demineralized Déminéralisée<br>Potable Potable | 31<br>1.0<br>1.0 | 1 |
| CHEMICALS<br>PRODUITS CHIMIQUES<br>VI | - | Surfuric Acid Acide sulfurique<br>Hydrochloric Acid Acide chlorhydrique<br>Phosphoric Acid Acide phosphorique | < 750<br>1.0 - 160<br>1.0 - 296 | 1.0 - 1.85 |

**Table 2 - Families of Liquids as related to Mass Flow Meters (Coriolis Meters)**

| Product Groups | Typical Products | Specific Gravity | Minimum Test Requirements to Cover Entire Subgroup |
|---|---|---|---|
| Normal Liquids | Water, Alcohols, Glycols, Water Mixes thereof, Agricultural Liquids, Fertilizers, Chemicals, Petroleum Solvents, Herbicides, and Suspensions | 0.7 to 2.5 | Test with one liquid having a density of 700 kg/m3 and with another liquid having a density of 2500 kg/m3.<br><br>**One product from this category will be used for durability testing.** |
| Compressed Liquids | Propane, Butane, Freon 11, Freon 12, Freon 22, $NH_3$, etc. | 0.3 to 0.68 | Test with one product having a density at any point within the range 300 kg/m3 to 6800 kg/m3.<br><br>**One product from this category will be used for durability testing.** |
| Cryogenic Liquids (BP 152°C) | Liquified Oxygen, Nitrogen, etc. | 0.07 to 1.4 | Test with one liquid having a density at any point within the range700 kg/m3 to 1400 kg/m3.<br><br>**One product from this category will be used for durability testing.** |
| Heated Products (above 50°C) | Bunker C, Asphalt, etc. | 0.8 to 1.2 | Test with one liquid having a density at any point within the range 800 kg/m3 to 1200 kg/m3.<br><br>**One product from this category will be used for durability testing.** |